
BIOLOGY

9700/51

Paper 5 Planning, Analysis and Evaluation

May/June 2017

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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PUBLISHED**Mark scheme abbreviations**

;	separates marking points
/	alternative answers for the same point
R	reject
A	accept (for answers correctly cued by the question, or guidance for examiners)
I	ignore (for answers that include irrelevant information that does not contradict the expected answer)
AW	alternative wording (where responses vary more than usual)
ora	or reverse argument (for answers which are written as the opposite to the expected answer)
<u>underline</u>	actual word given must be used by candidate (grammatical variants accepted)
max	indicates the maximum number of marks that can be given
ecf	error carried forward
mp	marking point (with relevant number)

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Question	Answer	Mark	Guidance
1(a)	<i>idea that</i> , results / they / it / test, could be affected by subject expectation / AW ;	1	A 'results' in terms of, heart beat / heart rate / pulse rate / reaction time AW I may affect results / fair test / <i>ref. to</i> reliable results unqualified
1(b)(i)	<i>independent variable</i> presence (or absence) of caffeine ; <i>dependent variables</i> heart rate and , reaction / response, time ;	2	A caffeine concentration / volume of caffeine A pulse rate / number of (heart) beats per minute A description of reaction time, e.g. time to press switch
1(b)(ii)	<i>max 8 of:</i> 1. <i>ref. to</i> having a large number of test subjects / AW ; 2. <i>ref. to</i> subjects / groups, have drinks with and without caffeine ; 3. <i>ref. to</i> description of method of making drinks indistinguishable or <i>ref. to</i> method that only, experimenter / student, can tell which is which ; <i>procedure</i> 4. <i>ref. to</i> test-subjects not, drinking / taking in, any caffeine (drink) for at least 5 hours before the test ; 5. <i>Idea that</i> each subject being tested in isolation / away from others (throughout the experiment) ; 6. <i>ref. to</i> subject, at rest / quiet, during test / after test / whilst having measurements (reaction time and heart rate) taken ; 7. <i>ref. to</i> taking measurements (of reaction time and heart rate) before giving the drink ;	8	1. if number stated, minimum of 10 with caffeine 2. A water / 0 mg caffeine / AW as decaffeinated drink

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Question	Answer	Mark	Guidance
	8. <i>ref. to</i> waiting (a minimum of) 45 minutes after giving the drink before measuring (the reaction time and heart rate) ; 9. <i>ref. to</i> giving the same volume of drink (to all subjects) ; 10. <i>ref. to</i> test subjects being, all caffeine / all non-caffeine, users ; 11. <i>ref. to</i> test subjects being, same / similar, age / mass / weight / fitness level / ethnicity / race ; 12. <i>idea of</i> standardised sex balance ; 13. calculate mean (for the measurements) ; 14. <i>ref. to idea that</i> health questionnaire / getting permission before testing / being aware of potential health risks or <i>ref. to</i> allowing test-subjects to stop if they feel unwell ;		8. A other stated times up to 2 hours / 120 minutes R around / about 45 minutes 9. A stated volumes, e.g. 100 – 350 cm³ A ‘a can’ / ‘a bottle’ / ‘a cup’ 10: I <i>ref to</i>. paired sampling 11. A same age range (e.g. 35–40) I <i>ref. to</i> health here (see mp 14) 12. A have all of one sex / equal numbers of one sex 13. I average A repeat three times and take a mean / AW 14. I low risk experiment R no risk A <i>idea of</i> excluding / being aware of, people with, caffeine or coffee or drink allergy / epilepsy (brought on by flashing lights) / heart conditions / neurological conditions / pregnancy / asthma I allergy / diabetes unqualified
1(c)(i)	(both sets of) data are continuous / data (are approximately) normally distribution / scatter graph or data or it suggests or shows a linear correlation / 5 or more paired observations ;	1	A interval data A relationship for correlation I trend / pattern
1(c)(ii)	there is a negative (linear) correlation / as caffeine concentration increases as reaction time decreases ;	1	A relationship for correlation I trend / pattern I qualification, e.g. strong / weak

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Question	Answer	Mark	Guidance
1(c)(iii)	subtract 2 from the (total) number of pairs of data ;	1	I number alone $df = n - 2 / 10 - 2 (= 8)$ A $n - 1 / 10 - 1 (= 9)$ or $n = 10$ or $(n - 1) + (n - 1) / 10 - 1 + 10 - 1 (= 18)$ A categories / samples for pairs of data I subjects
1(c)(iv)	1. use the probability table at 5% / 0.05 ; 2. compare the (calculated) r value / 0.722, to the critical value / 0.632 ; 3. significant, if / as, (calculated) r value / 0.722, is higher than critical value / ora ;	3	2. A table / tabulated, values as AW for critical values A <i>ref. to</i> higher / lower as evidence of comparison 3. A (less than) 5% probability / $P = (<) 0.05$, that the value is due to chance A 95% chance, that it is significant / that it is not due to chance A 'reject the null hypothesis' I <i>ref. to</i> the sign + or – R if <i>ref. to</i> 'expected v observed' / significant difference
1(d)	1. <i>idea that</i> only one person was tested for each of the concentrations ; 2. <i>idea that</i> a response could be, atypical / anomalous or people vary in their response (to caffeine) / an example of a possible variation in response ;	2	1. A only 10 subjects I not a large number of / not enough, subjects / AW 2. A <i>ref. to</i> subject 5 or 6 is anomalous I <i>ref. to</i> other experimental conditions not being controlled.
1(e)	<i>idea that</i> concentration of acetylcholine remains high (in synapses) so <i>idea that</i> reaction time is faster / (muscles) respond more quickly / response is more rapid / (post synaptic) neurones (supplying muscle) continue to be stimulated ;	1	A more acetylcholine present / acetylcholine, remains for longer / not broken down / increases / constantly secreted, so I reduces time for impulse to travel / impulse (s) travel faster / more synapses / AW

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Question	Answer	Mark	Guidance
2(a)(i)	<i>max 2 of:</i> 1. the number of times the populations were sampled ; 2. the number of locations within each site ; 3. the time of year that the populations were sampled ; 4. the day on which samples were taken (was the same) ; 5. <i>idea that</i> (size always) measured, as length / to nearest mm / in mm ; 6. (one / the same) species of beetle (counted) ;	2	1. populations sampled, 10 / the same number of, times 3. A populations sampled in the, same / warmest, (3) months / time of year / season I temperature
2(a)(ii)	1. trap / collect / sample / capture / AW, beetles, mark and release ; 2. re-trap beetles, and count / record the number of marked beetles, out of the total number recaptured ;	2	<i>max 1 for</i> 'mark, release, recapture' unqualified <i>2 marks for</i> mark, release, recapture and correct formula I incorrect / incomplete formula correct formula = <u>number in first sample × number in second sample</u> marked number in second sample

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Question	Answer						Mark	Guidance
2(b)(i)	site	A	B	C	D	E	2	
	population of beetles	10 792	11 314	18 426	15 224	17 650		
	number of large beetles	6520	6276	10 687	6432	6523		
	number of small beetles	4272	5038	<u>7739</u> ;	8792	11 127		
	percentage of large beetles	60	<u>55</u> ;	58	42	37		
2(b)(ii)	supports <i>idea that</i> as the, percentage / proportion, of large beetles is, less in higher temperatures / more in lower temperatures ;						1	A E hottest and has smallest , percentage / proportion of large beetles I any ref. to population size / number A partially or not supported if justified, e.g. <i>idea that</i> no clear pattern throughout range / ABC (BC / AB / AC) percentages similar but temperature varies A idea that no data relating to actual body size

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Question	Answer	Mark	Guidance
2(c)	<i>max 2 of:</i> 1. as temperature increases, proportion / percentage of, large beetles decreases ; ora 2. <i>idea that</i> (temperature) variations between A, B and C do not seem to affect size much as, percentage / proportion, of large beetles similar (60, 55, 58) ; 3. <i>ref. to</i> differences in annual (temperature) range related to, percentage / proportion of, large / small, beetles ; 4. <i>ref. to</i> average yearly temperature values or groups of values linked correctly to stated percentage or proportion of, large / small, beetles ; 5. <i>ref. to</i> lowest average temperature values or groups of values linked correctly to stated percentage or proportion of, large / small, beetles ; 6. <i>ref. to</i> highest average temperature values or groups of values linked correctly to stated percentage or proportion of, large / small, beetles ;	2	<i>I population size / number</i> 1. A size as equating to proportion or percentage, e.g. body size decreases as temperature increases for ora
2(d)	<i>must state what the aspect of climate <u>change</u> is considered e.g. warmer / cooler / <u>more</u> extreme / wetter / drier / windier / stormier, etc</i> one example of at least one possible climatic change and effect on population number or beetle size ;	1	<i>the effect must be possible in relation to the aspect of climate change quoted but does not need to be justified</i> e.g. A if global temperature rises / falls, would expect, smaller / larger / AW, populations A if global temperature rises would expect a higher, percentage / proportion, of <u>small</u> beetles in the population ora A global warming may result in an overall smaller body size A global warming could increase predators so reducing beetles